**UNIVERSITI TEKNOLOGI MALAYSIA**

**TEST 1**

**SEMESTER I 2018/2019**

**SUBJECT CODE : SCSR/SECR2043**

**SUBJECT TITLE : OPERATING SYSTEM**

**COURSE : SCSR / SCSV / SCSJ / SCSB / SCSP**

**DURATION : 9.00 – 10.00 PM (1 HOUR)**

**DATE : 27th NOVEMBER 2021 (Saturday)**

**VENUE : Online**

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**INSTRUCTIONS TO CANDIDATES:**

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|---|---|
| **Name** | |
| **IC No.** | |
| **Year/Course** | *** 1 / 2 / SCSR** |
| **Section** | |
| **Lecturer's Name** | |

## OBJECTIVE QUESTIONS [30 MARKS]

Chapter 1 : Choose 16 Questions from LOTS

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|---|---|---|
| 1. | Select the **FALSE** statement regarding the Operating System. | [1] |
| A.<br>B.<br>C.<br>D. | Act as a middleware in between user and computer hardware<br>Mapping the selected program to be executed in main memory<br>Manage memory, file and I/O devices<br>Act as a firmware in handling user and computer hardware | |
| 2. | Select the **TRUE** statement regarding the multicore CPU. | [1] |
| A.<br>B.<br>C.<br>D. | All multiprocessor systems are multicore system<br>Not all multiprocessor systems are multicore system<br>Consume high operating power compared to single-core chips<br>Avoid parallel computation to occur | |
| 3. | Which of the situations reflect the operation of a multiprocessor systems? | [1] |
| A.<br>B.<br><br>C.<br>D. | Open Web browser, word document and MP3 player simultaneously.<br>Allow to read the comment, get info related to the number of viewers, while going Live on Instagram.<br>Solve complex equation at one time.<br>Assist the gamers to have a better interface and response while playing games. | |
| 4. | System calls can be divided into few categories **EXCEPT**? | [1] |
| A.<br>B.<br>C.<br>D. | File manipulation<br>Process control<br>Communication<br>CPU Prevention | |
| 5. | Choose the **FALSE** statement regarding the responsibility of the Operating System in managing the process? | [1] |
| A.<br>B.<br>C.<br>D. | Create and delete the user and system processes.<br>Suspend and resume the processes.<br>Provide a mechanism to synchronize the processes.<br>Provide a defense system against internal and external attacks. | |
| 6. | Choose **TWO** features that uniquely identify the Batch, Interactive and Real-time Operating Systems. | [1] |
| I.<br>II.<br>III.<br>IV.<br><br>A.<br>B.<br>C.<br>D. | Response Time<br>Method of data entering the system<br>Location of data during the execution<br>Location of data after the execution<br><br>I and III<br>I and II<br>II and IV<br>III and IV | |
| 7. | Which of the following is **NOT** part of the privilege instructions? | [1] |

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| A.<br>B.<br>C.<br>D. | Change mode of execution<br>Modify entries in the device-information status table<br>Turn off all interrupts<br>Reading the status of processor | |
| 8. | Choose **TRUE** statement(s) regarding Bootstrap program. | [1] |
| I.<br>II.<br>III.<br>IV.<br><br>A.<br>B.<br>C.<br>D. | Reside in the firmware<br>Load the kernel into main memory<br>Initialize all aspect of the system<br>Acts as a middleware between user and resources<br><br>I and IV<br>I, II and III<br>II and III<br>III and IV | |
| 9. | Which of the following storages are **NOT** managed by the Operating System? | [1] |
| I.<br>II.<br>III.<br>IV.<br><br>A.<br>B.<br>C.<br>D. | Registers<br>Cache<br>Solid State Disk<br>Main Memory<br><br>I and II<br>I and III<br>II and III<br>III and IV | |
| 10. | Which of the following Operating Systems that can be used in medical imaging system? | [1] |
| A.<br>B.<br>C.<br>D. | Real-time operating system<br>Time-sharing operating system<br>Network operating system<br>Interactive system | |
| 11. | Choose the **FALSE** statement regarding the virtualization in computing environments. | [1] |
| A.<br>B.<br>C.<br>D. | Allow users to use multiple OSes without having multiple systems.<br>Test the applications without having multiple systems.<br>Run multiple OSes for each server machine in data server.<br>Speed up compile time in developing multi platform application. | |
| 12. | The clustered system has been widely used in the following systems **EXCEPT**? | [1] |
| A.<br>B.<br>C.<br>D. | Weather forecasting<br>Scientific computing<br>Super computing system<br>Payroll computing system | |
| 13 | Match the resource managements (**A – D**) that suitable to its Operating Sytem activity. | [4]<br>HOTS |

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| A | Provide a mechanism to synchronize the process. |
| B | Optimize CPU utilization and response time. |
| C | Perform disk scheduling. |
| D | Provide a defense system against internal and external attacks. |

**Answers provided:**
Process management
Memory management
Mass storage management
File management
I/O systems
Protection and security

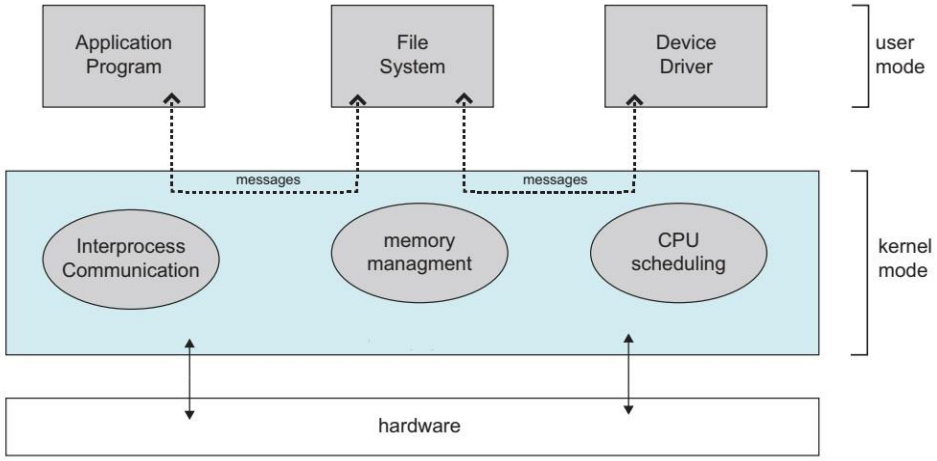
**ANSWER are:**
A- process management
B- mass storage management
C- protection and security
D- memory management

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| 14. | Operating System is an interface between _____ and _____ | [1] |
| A.<br>B.<br>C.<br>D. | User, Hardware<br>User, Applications<br>Applications, Hardware<br>User, CPU | |
| 15. | Which operating system is **NOT** an example of GUI systems. | [1] |
| A.<br>B.<br>C.<br>D. | MS-DOS<br>Windows<br>Android<br>Mac | |
| 16. | Which of the following is **NOT** a component of Operating Systems | [1] |
| A.<br>B.<br>C.<br>D. | Path<br>Shell<br>File<br>SystemKernel | |
| 17. | Which of the following is **NOT** an example of Multiprogramming Operating System? | [1] |
| A.<br>B.<br>C.<br>D. | MS-DOS<br>Mac OS X<br>Linux<br>Windows | |
| 18. | The computer system that allows more than one process execution at a time is known as: | [1] |
| I.<br>II. | Multiprocessing System<br>Single Core System | |

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| III.<br>IV.<br><br>A.<br>B.<br>C.<br>D. | Hyper Threading System<br>Single processing Systems<br><br>II and III<br>I and II<br>III and IV<br>I and III | |
| 19. | The response time is crucial in _____ Operating System. | [1] |
| A.<br>B.<br>C.<br>D. | real-time<br>batch<br>embedded<br>interactive | |
| 20. | What is the first program run on a computer when the computer start? | [1] |
| A.<br>B.<br>C.<br>D. | Bootstrap<br>DOS<br>System Program<br>Application Program | |
| 21. | Operating System can detect and correct errors. | [1] |
| A.<br>B. | No<br>Yes | |

Chapter 2: Choose 5 Questions from LOTS.

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| 22. | In the layered approach operating system structure _____. | [1] |
| A.<br>B.<br>C.<br>D. | bottom layer (layer *0*) is the hardware<br>highest layer (layer *N*) is the hardware<br>bottom layer (layer *0*) is the user interface<br>highest layer (layer *N*) is the user interface | |
| 23. | The command interpreter is also called the _____ | [1] |
| A.<br>B.<br>C.<br>D. | prompt<br>kernel<br>shell<br>command | |
| 24. | *These programs function is to create, modify the content of files stored and to search contents of files or perform transformations of the text.*<br><br>The statement refers to: | [1] |
| A.<br>B.<br>C.<br>D. | communications<br>background services<br>status information<br>file modification | |




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| 25. | Based on the given figure, what type of Operating System structure is indicated? | [1] |
| A. | Layered | |
| B. | Microkernel | |
| C. | Modules | |
| D. | Simple structure | |
| 26. | Following are the types of user interface of the Operating System **EXCEPT** | [1] |
| A. | batch interface | |
| B. | device interface | |
| C. | command line interface | |
| D. | graphic user interface | |
| 27. | `ioctl()`<br>`read()`<br>`write()`<br><br>Given are some of the UNIX system calls. The system calls can be grouped into: | [1] |
| A. | Device management | |
| B. | File management | |
| C. | Protection | |
| D. | Information maintenance | |
| 28. | Which of the following is **NOT** provided by Operating System Service? | [1] |
| A. | I/O operations | |
| B. | Accounting | |
| C. | Gaming | |
| D. | User interface | |

Chapter 3: Choose 5 Questions from LOTS and all HOTS Questions are compulsory.

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| 29. | Choose the **FALSE** statement describing a program. | [1] |
| A.<br>B.<br>C.<br>D. | Execution of program can be started via GUI mouse clicks.<br>An active program becomes process when executable file loaded into memory.<br>All the executable files that stored in on disk are categorized as an active program.<br>A program can be several processes. | |
| 30. | A process is more than a program code, which uses :<br>the__(A)___ in the memory that contains the global variables;<br>comprise a process ___(B) ___ which contains temporary data.<br><br>**ANSWER provided:**<br>Stack<br>Heap<br>Data section<br>Text<br>Program<br>shell<br><br>**ANSWER are:**<br>A- data section<br>B- stack | [1]<br>HOTS |
| A.<br>B.<br>C.<br>D. | stack<br>text<br>data section<br>heap | |
| 31. | Process to change states can be depicted by current activity of a process using _____. | [1] |
| A.<br>B.<br>C.<br>D. | Two-State Event<br>State Event Diagram<br>State Transition Diagram<br>Five-State Running | |
| 32. | The following statements are correct about process **EXCEPT**: | [1] |
| A.<br>B.<br>C.<br>D. | A process can be single threaded.<br>Process creation is light-weight.<br>A process can be multithreaded.<br>Process owns the resources. | |
| 33. | Which of the following are two operations that provided by the Inter-Process Communication? | [1] |
| A.<br>B.<br>C.<br>D. | write and delete message<br>delete and receive message<br>send and delete message<br>receive and send message | |

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| 34. | As processes enter the system, the processes migrate among the process scheduling queues. What are the queues involved?<br>___(A)___ contains a list of proceses in the system<br>___(B)___ is a list of proceses that residing in main memory and are ready and waiting to execute<br>___(C)___ is a list of processes waiting for the completion of I/O request<br><br>**ANSWER provided:**<br>Job queue<br>I/O queue<br>Ready queue<br>List queue<br>Program queue<br><br>Answers are:<br>A - Job queue<br>C - I/O queue<br>B - Ready queue | [3]<br>HOTS |
| 35. | Choose **TRUE** statement regarding the direct communication? | [1] |
| A.<br>B.<br>C.<br>D. | A communication link can be associated with *N* number of process (*N* = max. number of processes supported by system)<br>A communication link is associated with exactly two processes<br>Exactly *N/2* links exist between each pair of processes (*N* = max. number of processes supported by system)<br>Exactly two link exists between each pair of processes | |
| 36. | Currently CPU is executing Process Y.<br>Suddenly, I/O interrupt arises.<br>Process X is in ready state and coming to running state.<br>Then, context switching happens when _____ and _____.<br><br>**ANSWER provided:**<br>saving the state of Process X into $PCB_X$<br>loading the saved state of Process Y from $PCB_Y$<br>saving the state of Process Y into $PCB_Y$<br>loading the saved state of Process X from $PCB_X$<br><br>Answers are:<br>saving the state of Process Y into $PCB_Y$<br>loading the saved state of Process X from $PCB_X$ | [2]<br>HOTS |
| A.<br>B.<br>C.<br>D. | I and II<br>II and III<br>III and IV<br>I, II and III | |